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Winter Ice-Wave Modeling With WAVEWATCH III in Lake Erie

Haoguo Hu[1], Dan Titze[2], Ayumi Fujisaki-Manome[1,3], Bryan Mroczka[2], Jia Wang[2], Nathan Hawley[2], Sophie Orendorf[1], Kaitlyn Frank[1], and Steve Ruberg[2]

[1]Cooperative Institute for Great Lakes Research, University of Michigan, Ann Arbor, MI, USA, [2]NOAA, Great Lakes Environmental Research Laboratory (GLERL), Ann Arbor, MI, USA, [3]Climate and Space Sciences and Engineering, University of Michigan, Ann Arbor, MI, USA

Abstract Operational wave forecasts for the Great Lakes originate from the NOAA Great Lakes Waves Unstructured version 2 system. The model uses a simple ice blocking (IC0) parameterization for ice-wave damping, so ice-covered portions of the lakes are treated as land in the modeling system. Although simple and effective, the simple block can impede forecasting by eliminating wave forecast guidance from areas with thin or partial ice cover. We evaluate 12 ice-wave damping parameterizations within WAVEWATCH III (WW3, version 6.07.1) for Lake Erie, by comparing model results against wave observations made at several locations using moored acoustic wave and current profilers during the winters of 2010–2011 and 2012–2013. The comparisons show that the IC4M4 module performs the best among 12 ice modules with a root mean square error (RMSE) of 0.32–0.39 m and a root bias of −0.06 to −0.11 m, outperforming the existing IC0 parameterization (RMSE: 0.46–0.59 m; bias: −0.23 to −0.34 m) during the 2010–2011 analysis year. WW3 ice modules are mostly derived from measurements and studies of the Arctic and Antarctic Ocean. The dominant wave frequency is about 0.05∼0.10 Hz in the Arctic Ocean compared to 0.15∼0.2 Hz in the lake. Thus formulas built on frequency based on the studies from deep oceans may not be suitable for the shallow lakes because they cause too much damping. Although the IC4M4 ice module is from the study of the Antarctic Ocean , the wave attenuation formula based on incoming wave height is also suitable for Lake Erie.

Plain Language Summary Wave models are used to provide forecast guidance on wave conditions throughout the Great Lakes, improving safety for navigation and recreation. In the winter, waves on the Great Lakes are dampened by ice cover. To represent this in the wave model, the current NOAA Great Lakes operational wave forecast system uses a simple "ice blocking" scheme to simulate ice-induced wave damping, whereby portions of the lake with greater than 50% ice cover are treated as land in the model. In this scheme, no wave information is generated in these partially ice-covered areas, which can at times represent the majority of the lake surface. We evaluate 12 advanced ice-induced wave damping schemes within the wave model by comparing modeled wave heights to under-ice wave observations collected during the winters of 2010–2013. Of the methods tested, an empirical scheme ("IC4M4") with wave damping occurring proportional to wave height performed best, and improved upon the existing simple ice blocking scheme. When using this improved ice-induced wave damping scheme, wave predictions are generated throughout the entire model domain, thus benefiting forecasts by providing more accurate and comprehensive coverage of predicted Great Lakes wave conditions.

Key Points:

- The IC4M4 module performs the best for Lake Erie among the ice modules provided by WAVEWATCH III.
- Wave attenuation (by ice) formulas based on wave frequencies that are effective for deep oceans may not be suitable for shallow lakes.
- Wave attenuation (by ice) formulas based on wave heights are effective in both deep oceans and shallow lakes.

Correspondence to:

H. Hu and D. Titze,
hghu@umich.edu;
dan.titze@noaa.gov

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Author Contributions:

Conceptualization: Haoguo Hu
Data curation: Haoguo Hu, Dan Titze, Nathan Hawley, Sophie Orendorf, Kaitlyn Frank, Steve Ruberg
Formal analysis: Haoguo Hu, Dan Titze
Funding acquisition: Dan Titze, Ayumi Fujisaki-Manome, Jia Wang
Investigation: Haoguo Hu, Dan Titze
Methodology: Haoguo Hu, Dan Titze
Project administration: Dan Titze, Ayumi Fujisaki-Manome, Jia Wang
Resources: Ayumi Fujisaki-Manome, Jia Wang
Supervision: Dan Titze, Ayumi Fujisaki-Manome, Jia Wang
Validation: Haoguo Hu, Dan Titze
Visualization: Haoguo Hu, Dan Titze
Writing – original draft: Haoguo Hu

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1. Introduction

Physics-based wave modeling rapidly matured through the latter half of the 20th century, with advancements broadly categorized into three generations of wave models. First-generation wave models consisted of simple parameterizations of the relationships between wind, wave height, and wave period. These models have advanced to the current third-generation models in which four-wave nonlinear interaction and various other source terms are explicitly modeled in frequency-directional space, for example, WAM (WAMDI Group, 1988), SWAN (Booij et al., 1999) and WAVEWATCH III (WW3) (Alves et al., 2023; Tolman et al., 2002; WW3DG, 2019). For Lake Erie, one of the five Great Lakes, first- and second-generation physics-based wave models used simple parameterization to account for the evolution of wave height and period in response to wind forcing (Schwab et al., 1984), based on the relationships established by Donelan (1980). Third-generation wave models such as SWAN and MIKE21 SW (Sorensen et al., 2004), which explicitly represent three- and four-wave interactions,

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have subsequently been applied to Lake Erie (Moeini & Etemad-Shahidi, 2007). A coupled wave-hydrodynamic model based on the unstructured grid Finite Volume Community Ocean Model (FVCOM; Chen et al., 2013) and a third-generation wave model based on SWAN (Qi et al., 2009) was applied to Lake Erie to investigate wave climatology and inter-basin wave interactions (Niu & Xia, 2016). Alves et al. (2014) first implemented the third-generation WW3 model as the dynamical core of the National Oceanic and Atmospheric Administration's (NOAA) Great Lakes Waves operational forecast model. The WW3-based operational forecast system has since been updated to use an unstructured mesh and improved nearshore physics, with the most recent upgrade implemented by Abdolali et al. (2024) as Great Lakes Waves Unstructured version 2.0 (GLWUv2.0). In addition to these physics-based wave models, Hu et al. (2021) used data-driven deep learning models, including "eXtreme Gradient Boosting" (XGBoost) and "long short-term memory" (LSTM), to simulate waves in Lake Erie. These models outperformed WW3 in speed and accuracy at the locations tested, though its broader application is limited by the need for location-specific observational data sets to train the model.

Although the WW3-based GLWUv2.0 forecast system generally demonstrates strong skill during the ice-free season (Abdolali et al., 2024), studies seldom focus on validating wave models during the winter on Lake Erie or the other Great Lakes, largely due to a dearth of available observations. Validation of wave models in the Great Lakes is generally conducted by comparing model results to buoy-based observations of waves. Since the Great Lakes form ice cover each winter, buoys are recovered each fall and re-deployed in the spring to avoid damage from ice. This results in a seasonal bias, whereby there are typically no observations available for model validation from late fall through early spring, which is a period when strong winds and high waves are more common on the lakes.

In addition, ice observations on the Great Lakes needed to force the wave model during the winter are also limited, especially for a forecast model. Most information on ice cover in the Great Lakes comes from the North American Ice Service Great Lakes Daily Ice Charts (NAIS, 2023), which is produced once per day using available satellite imagery. Although these charts typically offer the best-available representation of ice conditions on the Great Lakes, they are limited by the coarse daily temporal resolution, their accuracy can be affected by the availability of satellite data such as visible imagery on cloudy days, and they do not regularly forecast ice conditions. Ice-wave modeling studies have become possible in recent years, with the development and application of the ice-lake model in the Great Lakes. Dynamic ice cover has been successfully simulated throughout the Great Lakes using a version of FVCOM modified for freshwater and internally coupled with an unstructured implementation of the Los Alamos Sea Ice Model (CICE) (Anderson et al., 2018; Gao et al., 2011; Hunke et al., 2015), though the ice outputs have not been evaluated as forcing for WW3. Bai et al. (2020) used FVCOM-CICE-SWAN to quantify wave energy loss with WW3 IC4 M1-7 parameterizations for the five Great Lakes implemented into the SWAN model. Ice cover simulated by the one-way coupled model significantly dampens significant wave height (SWH) as much as up to 80%–90%, depending on the amount of ice cover. Their practical hindcast application in 2011 with IC4M6 failed to reproduce the observed wave conditions with the opening of ice cover during the middle of the winter. The modeled SWH was consistently zero during the ice-covered period because the modeled ice cover did not melt as fast as observed in response to the synoptic warming.

The ice module used by the operational GLWUv2.0 model for ice-wave damping is a simple ice-blocking parameterization (IC0). The IC0 module treats the ice covered water as land in WW3, assuming zero wave energy and boundary conditions at ice edges are identical to boundary conditions at shore lines. The IC0 scheme is simple and effective for masking waves in a fully ice-covered system; however, its binary treatment of ice cover results in no wave damping in partially ice-covered areas below the ice concentration threshold, and a complete masking of waves above the threshold. This can hinder the process of operational wave forecasting within areas of partial or thin ice-cover, especially if the representation of ice cover is incorrect or offset. Furthermore, the abrupt and discontinuous changes in the model grid from water to "land" can result in unrealistic wave fields around the ice edge.

Here we evaluate alternative ice-wave damping parameterizations for WW3 in Lake Erie, using results from a loosely coupled FVCOM-CICE-WW3 model. We compare model results with under-ice wave observations from moored acoustic wave and current (AWAC) profilers during times when observations from surface buoys are not available. This work represents the first comprehensive evaluation of different ice-wave damping schemes within the Great Lakes using WW3. Improved ice-wave damping has the potential to offer improved forecast guidance during critical high-wind-wave seasons by better capturing the spatial distribution of the wave field in partially

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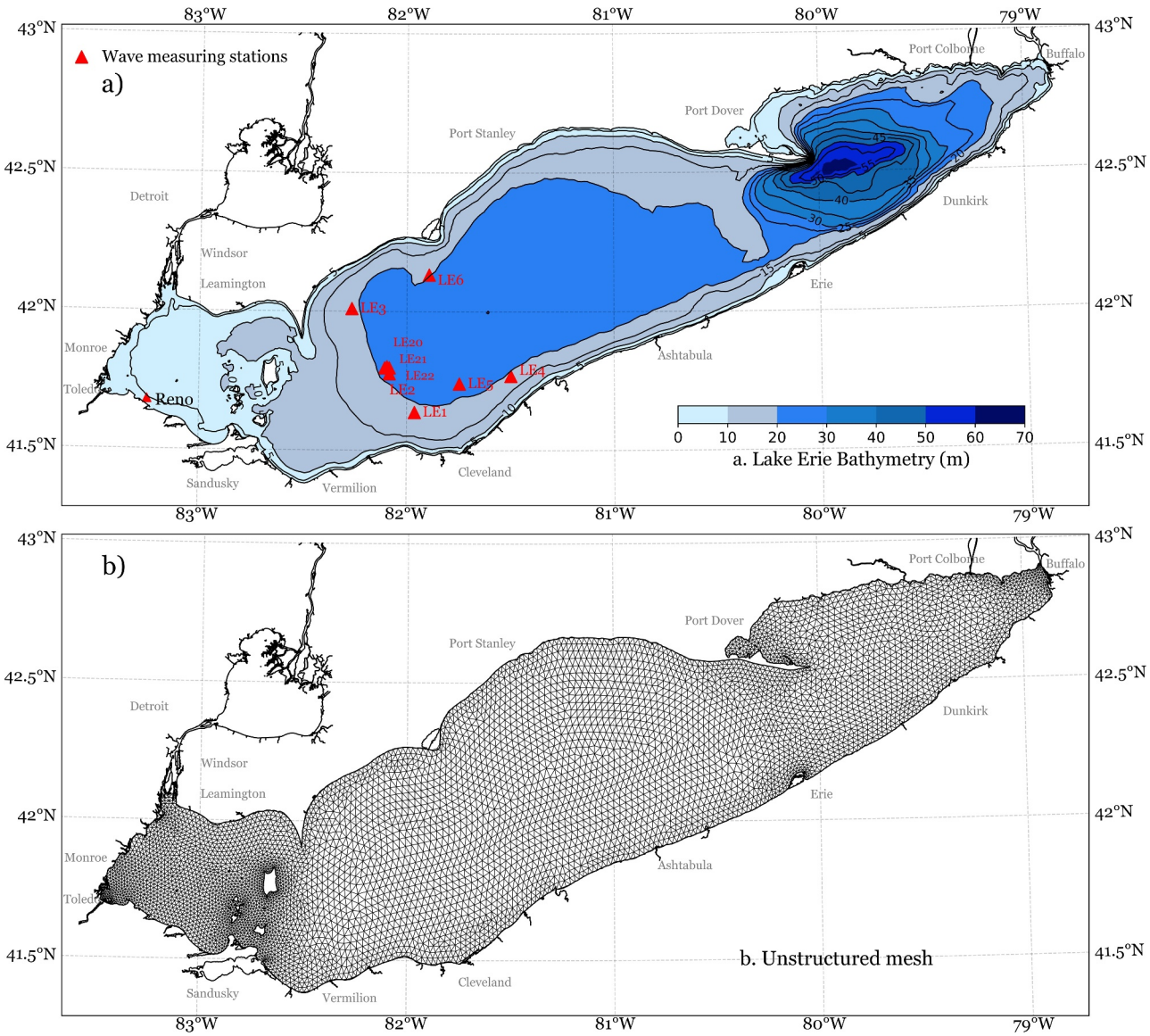


Figure 1. (a) Lake Erie bathymetry and locations of in situ measurement stations and (b) unstructured mesh of the wave model.

ice-covered portions of the lake, with benefit to commercial navigation, search and rescue, coastal resiliency, and recreation.

2. Methods

2.1. Model Domain and Configuration

Lake Erie is the southernmost of the Great Lakes, and it is also the shallowest, with an average depth of 19 m and a maximum depth of 64 m. It is relatively narrow, with a length of roughly 338 km and a width of roughly 92 km. Lake Erie can be broadly divided into three sub-basins based on bathymetric ridges, with a mean depth of 7.3 m in the western basin, 18.3 m in the central basin, and 24.4 m in the eastern basin (Figure 1a). Despite experiencing a moderately warmer climate than the other Great Lakes due to its more southerly location, Lake Erie typically forms ice each year, as its shallow water column cools comparatively quickly in the fall and winter (Wang et al., 2017, 2018). This is especially true of the shallow western basin of the lake, where landfast ice is historically common during the winter.

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The unstructured mesh used for the WW3 and FVCOM-CICE simulations is shown in Figure 1b, and is the same mesh used in the FVCOM-based NOAA Lake Erie operational forecast system (LEOFS; Kelley et al., 2018). The mesh is composed of 11,509 triangular elements and 6,106 nodes, with a horizontal resolution varying from approximately 350 m nearshore to 2.5 km offshore. Similar to the operational GLWUv2.0 model (Abdolali et al., 2024), this configuration concentrates the computational domain in shallower nearshore areas where wave dynamics are more influenced by interactions with the lake morphology and bathymetry, while maintaining adequate resolution in offshore areas to accurately simulate the distribution of waves in deeper portions of the lake. The model bathymetry is based on the data set hosted from the National Centers for Environmental Information (NCEI) (NGDC, 1999), and is referenced to the Great Lakes low water datum of 173.5 m. Water levels in the FVCOM-CICE vary based on hydrologic inputs and outputs to the lake, whereas the water level in WW3 is constant at the low water datum.

For the one-way coupled modeling system, FVCOM-CICE was run first to generate simulations of dynamic ice conditions. The model configuration used is the same as described by Anderson et al. (2018), which includes forcing for surface boundary conditions and major hydrologic inflows and outflows. The wave model was configured using WW3 version 6.07.1 with an explicit solver and the same physics packages as GLWUv2.0 (Abdolali et al., 2024), but with varying the ice damping physics package. The hourly ice concentration output from the FVCOM-CICE model was used as a forcing field in the WW3 model, and there was no feedback from the WW3 model to the FVCOM-CICE model. Wind and ice concentration are the only forcing inputs to the WW3 model.

The ice analysis data set (Section 2.2) was also considered as a source of ice concentration data for forcing WW3. However, the hourly temporal resolution of the FVCOM-CICE data is advantageous because Lake Erie can experience large intraday changes and movements in ice cover that would not be resolved by the daily ice analysis data set. More critically, to achieve the ultimate goal of this work and implement improved ice damping in a realtime forecast model, forecast ice cover from a model would be the only viable source of future ice conditions, which is needed to drive the wave forecast model.

For the hindcast configuration, surface boundary conditions, including those for determining surface heat flux in the FVCOM-CICE model and the 10-m wind speed for both FVCOM-ICE and WW3, are from interpolated observations (data from buoys and coastal stations) in simulations for 2010–2013 winters using the sources and methods described by Schwab and Bedford (1994). To evaluate the skill of the loosely coupled modeling system in a forecast setting, we also configured the modeling system in a pseudo-real-time mode for the winter of 2022–2023. The pseudo-real-time system consists of a nowcast, which is forced by NOAA operational high-resolution rapid refresh (HRRR; Dowell et al., 2022; James et al., 2022), as well as a 3-day forecast, which is forced by HRRR for forecast hours 0–48 hr and the NOAA operational global forecasting system model (GFS; NCEI, 2023) for forecast hours 49–72.

The model configuration used WW3 version 6.07.1. The model settings were chosen to match those of the well-validated NOAA GLWUv2.0 operational model for the Great Lakes as our baseline configuration (Abdolali et al., 2024), with the ice-wave damping scheme being varied in our experiments. The simulations were run on a supercomputer allocated with 49 CPUs. For a 5-month (1 Dec 2010 to 1 May 2011) case, the maximum run time was for IC4M6 (6 hr 54 min), and the minimum running time was for IC1 (4 hr 27 min).

2.2. Observational Data

To evaluate the performance of WW3 with ice, waves and ice thickness from mooring observations are used for comparison. In situ observations of wave height and ice thickness were collected for the winters of 2010–2011 through 2012–2013 in the central basin of the lake; however, because little to no ice formed in the open waters of Lake Erie during the warm 2011–2012 winter, only the 2010–2011 and 2012–2013 winters are considered in this study. A list of observations considered in this study is presented in Table 1. Detailed descriptions of the in situ ice and wave measurements for the 2010–2011 season can be found in Hawley et al. (2018), who conducted the 2010–2013 field campaign. Winter wave data from one additional station, Reno Beach, was also used in our analysis (GLERL, 2023a).

Significant wave heights were measured using Nortek acoustic wave and current profilers (AWAC) at three stations in Lake Erie (LE4-6) during the winters of 2010–2011, and one station (LE21) during the winter of

Table 1
Description of In-Situ Observation Data Sets from Lake Erie Used in This Study

Season	Station	Latitude	Longitude	Instruments
2010–2011	LE1	41.637°N	81.962°W	SWIPS
	LE2	41.776°N	82.084°W	SWIPS
	LE3	42.010°N	82.266°W	SWIPS
	LE4	41.768°N	81.498°W	SWIPS, AWAC
	LE5	41.740°N	81.746°W	AWAC
	LE6	42.132°N	81.893°W	AWAC
2012–13	LE20	41.801°N	82.095°W	SWIPS
	LE21	41.797°N	82.105°W	SWIPS, AWAC
	LE22	41.797°N	82.084°W	SWIPS
2022–23	Reno Beach	41.680°N	83.250°W	AWAC

2012–2013 (Figure 1a). The measurement accuracy for AWAC is 1% of the measured SWH, and 1 cm for resolution and a lower cutoff limit of 0.9 s (0.5 s) wave period for SWH measurements at 20 m (5 m) deployment depth. The 2010–2013 observation stations all have deployment depths of approximately 20 m (±2 m), whereas the 2022–2023 Reno Beach station has a depth of approximately 6 m. AWACs produce error codes to indicate when data points may be unreliable, and all data points with such error codes were removed from the 2010–2013 observational data sets to ensure data quality. For all data sets, data gaps of up to 6 hr were patched through linear interpolation, and larger data gaps were treated as missing values.

Ice thickness was also measured during the winters of 2010–2011 and 2012–2013 using a shallow water ice profiling sonar (SWIPS) manufactured by ASL Environmental Sciences. The SWIPS units were mounted 2–3 m above the lake bottom, supported by subsurface floats, and emitted a single vertical acoustic beam at 546 kHz to measure the range to the bottom of any ice present. In 2010–2011, SWIPS units were deployed at stations LE1, LE2, LE3, and LE4, and in 2012–2013 SWIPS units were deployed at stations LE20, LE21, and LE22 (Table 1; Figure 1a). SWIPS units were also deployed during the 2011–2012 season; however, because little to no ice cover formed at the open water locations, 2011–2012 measurements are not further discussed. The SWIPS units were configured to make range measurements each second, and measurements of water pressure, water temperature, pitch, and roll every 10 s. The temporal resolution of the measurements was 10 min, but was averaged to hourly for the analyses conducted herein. Due to noise in the measurements, some negative ice thickness values are present, and any negative values after hourly averaging were set to zero.

In addition to in situ observations, data sets depicting spatial ice coverage on the Great Lakes have been collected and maintained since 1973 by Canadian Ice Service, U.S. National Ice Center, and the NOAA Great Lakes Environmental Research Laboratory (GLERL). These ice data sets are produced based on analysis primarily of satellite imagery, but also included ship-based observations, or modeled conditions when beneficial. GLERL reprocessed the Great Lakes ice cover database to generate daily gridded data (1.8 km resolution) using a re-project method with nearest neighbor search for spatial interpolation and linear interpolation with categorization for temporal interpolation (Wang et al., 2017, 2018; Yang et al., 2020). Here, the GLERL reprocessed ice concentration data set (henceforth "ice analysis") is used for model-data comparison.

2.3. Ice Cover Validation

In general, the Lake Erie FVCOM-CICE model results have been previously shown to have general agreement with the ice analysis data set, though the model can misrepresent the timing of rapid freezing and thawing periods and can produce significantly too much ice during low-ice years (Anderson et al., 2018). Although the ice analysis is the best available representation of observed lake-wide surface conditions during the simulation periods, there is some uncertainty in the data set, as the quality of data used to produce the analysis varies due to the number of satellite passes on a given day, the presence of clouds when analyzing visual spectrum imagery, and it is subject to manual interpretation and synthesis of the underlying data sets. The availability of the SWIPS data provides a rare opportunity to compare modeled ice conditions to the ice analysis data set in the context of in situ observations.

Because SWIPSs are measuring ice thickness at a discrete point and ice analysis and modeled ice concentration data sets used in this study represent areal concentration across the domain, the data sets were compared using a binary presence or absence metric. Ice was said to be present in the SWIPS observations when hourly averaged ice thickness was at least 5 cm, which is the approximate detection limit of the SWIPS instruments (Hawley et al., 2018); ice was said to be present in the ice analysis and model output when areal concentration was at least 50%, which is the same threshold used by the WW3 model. Although this is not a direct comparison of like parameters, it serves to provide a coarse assessment of both the ice analysis and modeled ice cover as forcing data sources for wave damping. To quantify the performance, daily averaged model output and SWIPS data were compared to the daily ice analysis data set. Metrics for time-percent accurate, time-percent overicing, and time-percent undericing were calculated for both the model output and ice analysis data. Time-percentage accurate is the percentage of time that presence or absence of ice in the data set matches the in situ observations, time-percent

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overicing is the percentage of time that the data set shows ice when the in situ observations do not, and time-percent undericing is the percentage of time that the data set shows absence of ice when it is present in the in situ observations.

$$\text{Time Percent Accurate} = \frac{n_{\text{accurate}}}{n_{\text{total}}} * 100$$

$$\text{Time Percent Under Ice} = \frac{n_{\text{underIce}}}{n_{\text{total}}} * 100$$

$$\text{Time Percent Over Ice} = \frac{n_{\text{overIce}}}{n_{\text{total}}} * 100$$

where n_{total} is the total number of days in the comparison, n_{accurate} is the number of days where the result matches observations, n_{underice} is the number of days where the result shows ice absent when the observations show ice present, and n_{overice} is the number of days when the result shows ice present when the observations show ice absent.

This analysis is intended to provide a high-level comparison of the extent to which the modeled and ice analysis data sets agree with in situ observations at the open water locations examined in this study. Because accurate ice-wave damping inherently relies on the availability of accurate ice forcing data, understanding the extent to which results are influenced by the skill of the forcing as opposed to the skill of the parameterization is critical for identifying areas of future improvement in the coupled ice-wave model system.

2.4. Ice-Wave Damping Skill Assessment

We evaluate 12 different ice damping parameterizations in our simulations (Table 2), as well as an undamped case. Values of tunable parameters were chosen using values suggested in the literature or regression tests for empirical parameters, and using values representative of the Lake Erie system for parameters with a physical meaning, such as ice thickness and water depth. Simulations were run from December through April for each of the winter seasons evaluated, and hourly results were extracted and compared at locations with available in situ observations. Performance of the model cases was evaluated by comparing the hourly time-series of model output to observations using root mean square error (RMSE) and bias skill statistics, calculated as follows:

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n} \left(\hat{x}_i - x_i\right)^2}{n}}$$

$$\text{Bias} = \frac{\sum_{i=1}^{n} \left(\hat{x}_i - x_i\right)}{n}$$

where n is the number of data points, $\hat{x}$ is the modeled value, and x is the observed value. RMSE was chosen to quantify the accuracy of the model, whereas bias was chosen to quantify the tendency of the model toward overdamping or underdamping.

Skill assessment of the ice-wave damping modules focuses on the 2010–2011 winter season, which has moderate ice cover and in situ wave observations from three stations. Skill assessment is also conducted against observations from the 2012–2013 winter for the top-performing ice-wave damping parameterization identified through the 2010–2011 analyses, and in a pseudo-forecast mode for the 2022–2023 winter season.

3. Results and Discussion

3.1. Ice Cover Validation

Although not the main focus of this work, ice data sets were closely analyzed as part of this study, as accurate ice forcing data is critical for producing better ice-wave damping. Time-series of ice concentration from modeled and ice analysis data sets are shown alongside in situ SWIPS ice thickness observations in Figure 2. There was a shorter ice season in 2012–2013, with SWIPS ice thickness measurements showing ice present from mid to late January through mid to late March. It should be noted that while there were three SWIPS stations in 2012–2013,

Table 2
Description of Ice-Wave Damping Configurations Evaluated

	Description	Attenuation coefficient empirical relationship	Values of tunable parameters used	References
IC0	Simple ice blocking	N/A (see references for description)	N/A	WW3DG (2019)
IC1	Simple ice damping	$\alpha = 2\,C_1$	C_1 (damping coefficient, m^{-1}): 2.0×10^{-5}	Rogers and Orzech (2013), WW3DG (2019)
IC2	Viscoelastic damping with ice modeled as continuous thin elastic plate	N/A (see references for description)	C_1 (ice thickness, m): 0.15 C_2 (eddy viscosity, $m^2\ s^{-1}$): 1.0×10^{-5} C_5 (water depth, m): 20	Rogers and Orzech (2013), WW3DG (2019)
IC3	Viscoelastic damping with ice modeled as frazil ice floes	N/A (see references for description)	C_1 (ice thickness, m): 0.15 C_2 (effective viscosity of ice, $m^2\ s^{-1}$): 3 C_3 (ice density, kg m^{-3}): 917 C_4 (effective shear modulus of ice, Pa): 0	Wang and Shen (2010), WW3DG (2019)
IC4M1	Empirical exponential damping as a function of wave period (T), with higher damping for smaller-period waves	$\alpha = e^{\left(\frac{-2\pi C_1}{T} - C_2\right)}$	C_1, C_2 (empirical): 0.18, 7.3	Collins and Rogers (2017), Wadhams et al. (1988), WW3DG (2019)
IC4M2	Empirical polynomial-fit damping as a function of wave period (T), designed to be flexible	$\alpha = C_1 + C_2\left(\frac{T}{2\pi}\right) + C_3\left(\frac{T}{2\pi}\right)^2 + C_4\left(\frac{T}{2\pi}\right)^3 + C_5\left(\frac{T}{2\pi}\right)^4$	C_1–C_5 (empirical): 0, 0, 2.12×10^{-3}, 0, 4.59×10^{-2}, 1.1	Collins and Rogers (2017), Meylan et al. (2014), WW3DG (2019)
IC4M3	Empirical quadratic decay as a function of wave period (T) and ice thickness, with higher attenuation for thicker ice and smaller-period waves	$\alpha = exp[-3.203 + 2.058\,C_1 - 0.9375\,T - 0.4269\,C_1^2 + 0.1566\,C_1 T + 0.0006\,T^2]$	C_1 (ice thickness, m): 0.15	Collins and Rogers (2017), Horvat and Tziperman (2015) WW3DG (2019)
IC4M4	Empirical damping as a step function of significant wave height (Hs), with linear damping for Hs ≤ 3 m and capped damping for Hs > 3 m	$\alpha = \begin{cases} 2\,C_1 & \text{for } H_s \le 3\text{ m} \\ 2\,C_2\,H_s^{-1} & \text{for } H_s > 3\text{ m} \end{cases}$	C_1, C_2 (empirical): 5.35×10^{-6}, 16.05×10^{-6}	Collins and Rogers (2017), Kohout et al. (2014), WW3DG (2019)
IC4M5	Empirical damping as a step function of wave period (T), with four user-defined steps and damping coefficients	$\alpha = \begin{cases} 2\,C_1 & \text{for } T^{-1} < C_5 \\ 2\,C_2 & \text{for } C_5 \le T^{-1} < C_6 \\ 2\,C_3 & \text{for } C_6 \le T^{-1} < C_7 \\ 2\,C_4 & \text{for } T^{-1} \ge C_7 \end{cases}$	C_1–C_4 (damping coefficients, m^{-1}): 5.0×10^{-6}, 7.0×10^{-6}, 15.0×10^{-6}, 100.0×10^{-6} C_5–C_7 (step bounds, Hz): 0.10, 0.12, 0.16	Collins and Rogers (2017), WW3DG (2019)
IC4M6	Empirical damping as a step function of wave period (T), with up to ten user-defined steps and damping coefficients	$\alpha = \begin{cases} 2\,C_{A1} & \text{for } T^{-1} < C_{B1} \\ 2\,C_{A2} & \text{for } C_{B1} \le T^{-1} < C_{B2} \\ \cdots & \\ 2\,C_{A9} & \text{for } C_{B8} \le T^{-1} < C_{B9} \\ 2\,C_{A10} & \text{for } C_{B9} \le T^{-1} < C_{B10} \end{cases}$	C_{A1}–C_{A10} (damping coefficients, m^{-1}): 5.0×10^{-6}, 7.0×10^{-6}, 1.5×10^{-5}, 1.0×10^{-4}, 0, 0, 0, 0, 0, 0 C_{B1}–C_{B10} (step bounds, Hz): 0.10, 0.12, 0.16, 99.0, 0, 0, 0, 0, 0, 0	Collins and Rogers (2017), WW3DG (2019)
IC4M7	Empirical damping as a function of wave period (T) and ice thickness	$\alpha = 0.2T^{-2.13}C_1$	C_1 (ice thickness, m): 0.15	Doble et al. (2015), WW3DG (2019)
IC5	Viscoelastic damping with ice modeled as thin elastic plate restricted to one dimension	N/A (see references for description)	C_1 (ice thickness, m): 0.15 C_2 (effective viscosity, $m^2\ s^{-1}$): 1.6×10^7 C_3 (ice density, kg m^{-3}): 917 C_4 (effective shear modulus, Pa): 4×10^{12}	Mosig et al. (2015), WW3DG (2019)

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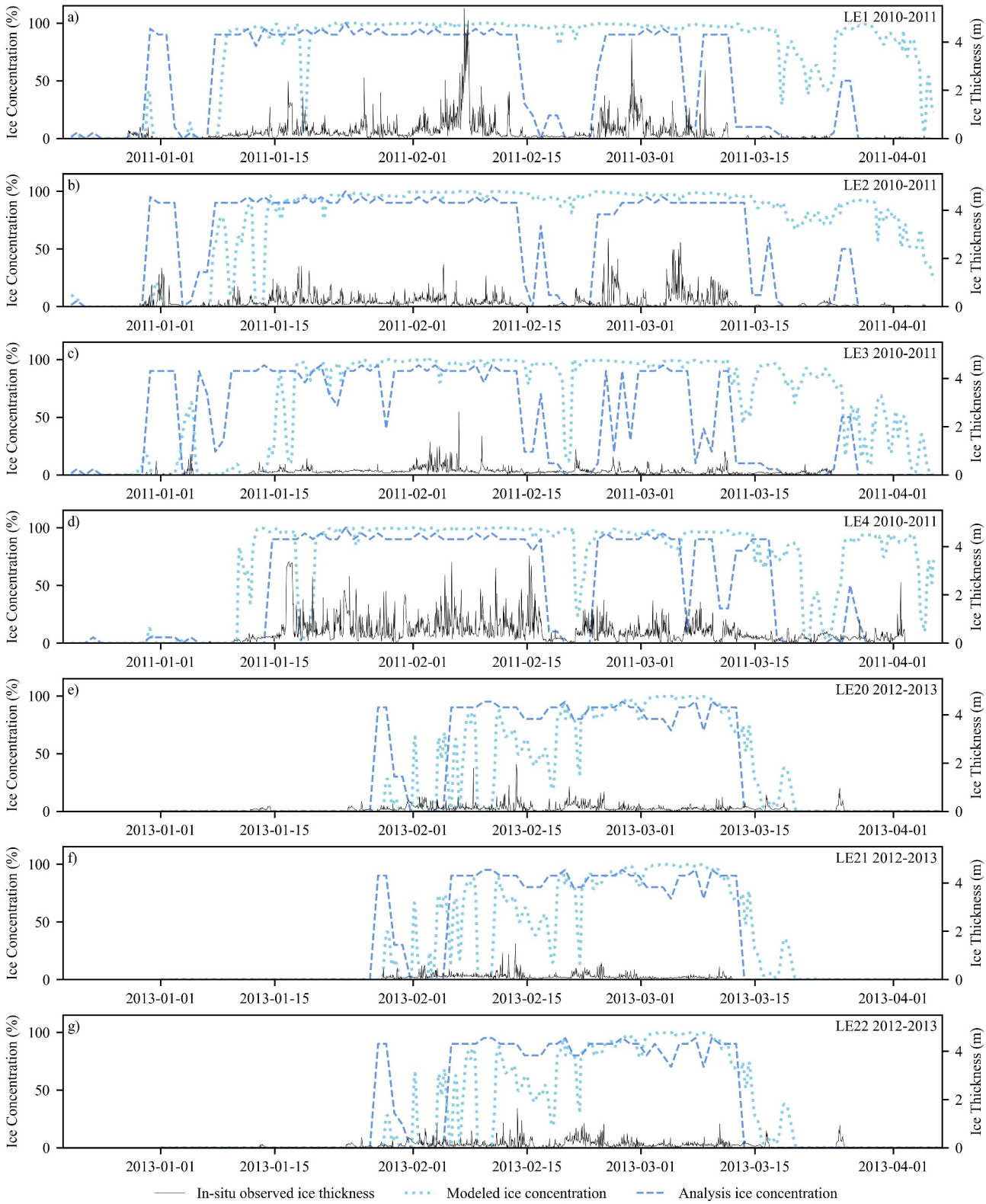


Figure 2. Ice comparisons at panels (a–d) LE1-4 in 2010–2011 and (e–g) LE20-22 in 2012-13. In situ ice thickness observations are shown in solid black (right *y*-axis), analyzed satellite ice concentration in dashed dark blue, and modeled ice concentration in dotted light blue (left *y*-axis).

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they were spaced in relatively close proximity to one another, so their distribution of measurements is more representative of localized spatial variability, whereas the distribution of 2010–2011 measurements represents a larger portion of Lake Erie's central basin. Although ice thicknesses are typically in the range of approximately 10–30 cm at most stations, spikes in ice thickness exceeding 1 m were observed at all stations, and exceeded 5 m at station LE1 in the 2010–2011 season. These spikes can be attributed to passing ice keels, as a result of ice ridging against the shoreline and drifting offshore.

Consistent with the conclusions of Anderson et al. (2018), the greatest differences between modeled and ice analysis data sets typically occur at the beginning and end of the ice season, which are characterized by rapid ice formation and ice melting. The ice analysis tends to show greater ice cover earlier in the season than the model results, and modeled ice consistently melts later in the season than is shown in the ice charts. Comparison of these timeframes to those that can be deduced from in situ ice thickness measurements are mixed, for both the onset and offset of ice cover. In 2010–2011 at station LE4, the earlier ice formation in the model compares better the timing of ice presence in the in situ observations than does the ice analysis, and at station LE21 in 2012-13 the later formation of ice in the model compares better to the timing of first ice presence in in situ observations than the ice analysis. However, for most other stations and years, the ice analysis representation of ice earlier in the season is more consistent with the times that ice was first observed in the in situ observations. At many stations and years, there is some degree of ice thickness detected by the SWIPS when ice has not completely melted in the model but the ice analysis depicts ice-free conditions; however, even in such cases, ice often remains in the model after it is no longer detected in the in situ observations. Additionally, at some of these stations, such as LE2 and especially LE1 in 2010–2011, the late-season ice thickness as measured by the SWIPS is thin and sporadic in nature. At LE21, both the ice analysis and model show ice present beyond when it was detected in the in situ observations.

The general tendency of later offset and to a lesser extent later onset of ice cover in the model compared to the ice analysis and in situ ice thickness observations suggests that modeled ice growth and decay may be responding too slowly to external forcing. This is evident not only in the initial growth and final melting stages of the ice season, but also through some episodic events. For example, during the 2010–2011 season, there is a period of low ice thickness observed at stations LE1, LE2, and LE4 between approximately 15 and 23 February, with the exact timing varying by station. There is a drop in ice concentrations in the ice charts at these stations during the event that roughly corresponds to the observations, whereas the model results show only a minimal or brief reduction in ice. However, while this example helps demonstrate the tendency of modeled ice growth and decline to respond too slowly to changing conditions, presence of ice cover in the ice analysis data set does not consistently align with presence of ice in the in situ observations. In the same event, the ice analysis at station LE3 shows a total decline of ice during the aforementioned February 2011 event that is not observed in the ice thickness event, and there are other such instances throughout the time-series.

Table 3 presents results from the simple presence and absence validation of the ice analysis and ice model data sets against the moored in situ ice thickness observations. As with the more qualitative discussion above, results are mixed. The ice analysis data set had a higher accuracy metric for LE1 and LE2 in 2010–2011, and for all three stations in 2012–2013, whereas the model had a higher accuracy for stations LE3 and LE4 in 2010–2011. The model bias was toward overicing in the 2010–2011 season, and toward undericing in the 2012–2013 season, suggesting that the bias characteristics of the model may vary from year to year based on conditions. Overall, despite the marginally higher accuracy metric of the ice analysis data set compared to the model, the validation metrics remain reasonably close between the two data sources. Therefore, the benefits of using the model data for this one-way coupled system, such as higher temporal resolution and availability of forecast data, do not come at a great cost compared to the ice analysis data sets, which remain the best available observational data set of spatial ice distribution for the domain. Data assimilation of the ice analysis into the model could provide a means for further improving the model, particularly at the beginning and end of the ice season when the greatest discrepancies are noted. In addition, identifying improvements to the model physics such that it responds more rapidly to external conditions may further improve the coupled modeling system. For example, a two-way-coupled ice-wave system that accounts for processes such as ice breakage due to waves may help accelerate ice melting in the spring, thus mitigating the delayed ice melting.

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Table 3
Ice Presence and Absence Evaluation from Modeled and Analyzed Results

Season	Station	Data set	Time-percent accurate (%)	Time-percent overice (%)	Time-percent underice (%)
2010–2011	LE1	Model	72.4	21.9	5.7
		Analysis	83.8	2.9	13.3
	LE2	Model	73.3	18.1	8.6
		Analysis	80.0	3.8	16.2
	LE3	Model	91.4	3.8	4.8
		Analysis	67.6	7.6	24.8
	LE4	Model	91.4	1.0	7.6
		Analysis	71.4	0.0	28.6
2012–2013	LE20	Model	74.7	0.0	25.3
		Analysis	77.8	2.0	20.2
	LE21	Model	81.8	3.0	15.2
		Analysis	90.9	2.0	7.1
	LE22	Model	78.8	0.0	21.2
		Analysis	81.8	2.0	16.2

3.2. Ice-Wave Damping Skill Assessment

Results from the twelve wave-ice damping parameterizations, as well as the undamped case, are presented in Figure 3 for station LE4 during the 2010–2011 winter season, with corresponding skill statistics summarized in Table 4. LE4 has the best temporal coverage for in situ observations of the three available stations during the 2010–2011 season, and represents a relatively nearshore location close to the population center of Cleveland OH, where coastal waves conditions may have societal impact. Unsurprisingly, results from all scenarios are comparable early in the season, prior to localized ice formation. However, some small differences are noted even in early January, due to upwind damping of waves as a result of early ice formation in the western basin of the lake, which was mostly ice covered by mid December (GLERL, 2023b; Yang et al., 2020). During the mid-January to mid-April portion of the winter season, when ice was present immediately surrounding the LE4 station, there werre substantial differences in modeled wave heights under the different ice-wave damping parameterizations. In situ observations show a general reduction in wave height during the ice-covered period, though occasional events producing waves in excess of 1 m occurred.

The undamped scenario (Figure 3a) demonstrates the need for ice-wave damping in the modeling system. There is a clear high bias (+0.15 m) in modeled significant wave height when ice-wave damping is not accounted for. The differences are largely episodic in nature, whereby there are discrete episodes of high waves in the undamped model that were not observed through the in situ measurements. However, there are also episodes of higher waves captured in the in situ observations that the undamped model is doing better at capturing than the damped models, most notable on 19 February 2011 when observed wave heights exceeded 2 m. This indicates that waves are at times overdamped in the damped scenarios, likely due to a combination of overly aggressive ice damping schemes and, as previously discussed, an imperfect representation of ice cover in the FVCOM-CICE model. In particular, the ice analysis data set showed a prolonged absence of ice cover at the LE4 location around the time of this wave event, whereas the FVCOM-CICE model simulates a much briefer drop in ice cover.

Under the existing IC0 simple blocking parameterization, wave heights are not modeled at the LE4 location for the majority of the winter season, due to the treatment of areas of the lake with greater than 50% ice coverage as land under this scheme (Figure 3b). This represents the most aggressive method of damping. Notable in the IC0 method is the abrupt spike in significant wave height on 11 January 2011, peaking at 4.6 m, resulting from the simplified treatment of ice cover in this configuration. The IC0 parameterization's treatment of the ice edge as land results in an abruptly changing lake morphology even in deep offshore regions, and such abrupt changes in the modeled lake boundary can produce these unrealistic spikes in wave heights along the ice edge. These spikes are not present in the observations or in any of the other model scenarios.

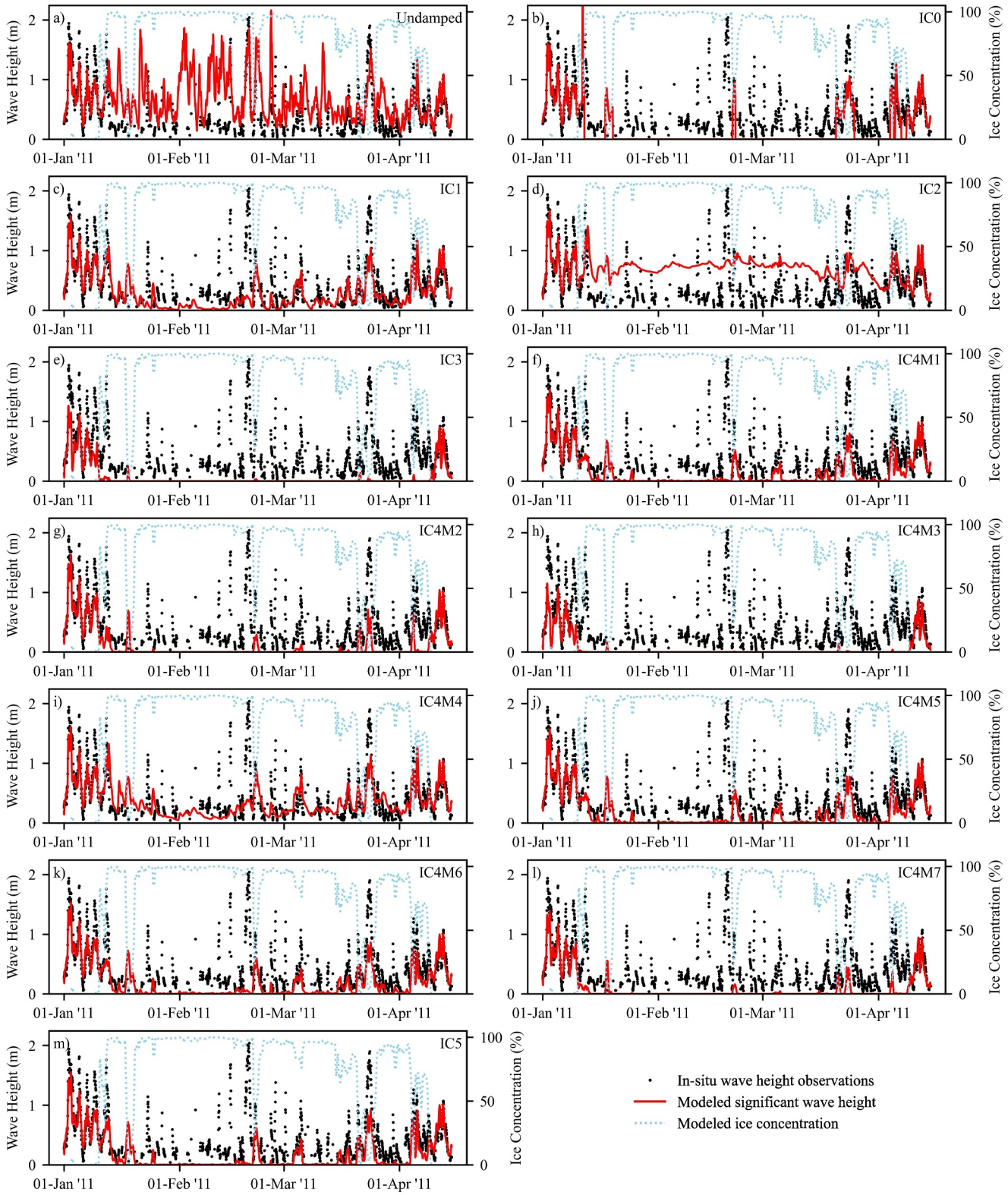


Figure 3. Comparison of modeled significant wave height at LE4 to in situ observations for cases using different ice-wave damping parameterizations. In situ measurements of significant wave height are shown as black dots, modeled wave heights are shown as a red line. Modeled ice cover is shown as a dotted light blue line for context.

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Table 4
Skill Statistics for Ice-Wave Damping Cases, Calculated by Comparing Modeled Significant Wave Height to In-Situ Observed Significant Wave Height From 1 January to 15 April 2011

	LE4 (n = 1,482)		LE5 (n = 1,080)		LE6 (n = 637)	
	RMSE	Bias	RMSE	Bias	RMSE	Bias
Undamped	0.37	+0.15	0.30	+0.06	0.27	+0.11
IC0	0.46	−0.23	0.59	−0.34	0.54	−0.23
IC1	0.34	−0.13	0.43	−0.19	0.43	−0.12
IC2	0.41	0.19	0.44	+0.18	0.34	+0.12
IC3	0.54	−0.38	0.68	−0.52	0.59	−0.40
IC4M1	0.42	−0.26	0.56	−0.37	0.51	−0.23
IC4M2	0.47	−0.31	0.63	−0.45	0.55	−0.28
IC4M3	0.55	−0.39	0.69	−0.52	0.61	−0.43
IC4M4	0.32	−0.06	0.38	−0.11	0.39	−0.08
IC4M5	0.41	−0.24	0.55	−0.34	0.50	−0.21
IC4M6	0.39	−0.22	0.52	−0.32	0.49	−0.21
IC4M7	0.49	−0.34	0.66	−0.48	0.56	−0.32
IC5	0.40	−0.23	0.53	−0.32	0.50	−0.21

The scenarios with more complex ice-wave damping parameterizations show a mixture of results. Of the parameterizations tested, the empirical IC4M4 parameterization demonstrates the strongest skill at this location during 2010–2011, with a RMSE of 0.32 m and a bias of −0.06 m. The IC4M4 parameterization is based on the relationship of Kohout et al. (2014) for the marginal ice zone of the Antarctic, where they found wave damping increases linearly with wave height up to 3 m, at which point the attenuation is capped. The temperate location of the Great Lakes in combination with their size results in a dynamic and partially ice-covered environment (GLERL, 2023b; Wang et al., 2017, 2018), similar in nature to the marginal ice zone of the Southern Ocean for which the scheme was developed.

The IC1 simple ice damping parameterization also performs reasonably well, with a RMSE of 0.34 m and a bias of −0.13 m. Results from each of these cases show a muted response to wind forcing during the ice-covered portion of the season, while still capturing, though generally underpredicting, some of the observed wave events. The IC1 module allows the user to specify a uniform attenuation coefficient. The coefficient is not a function of wave or ice conditions, and instead depends on the user's experiences. After deliberately tuning, one may gain comparable results to IC4M4, but the wave-correlated scheme of the IC4M4 and better skill using suggested parameter values seem to produce superior results in the Lake Erie domain.

Despite having middling skill statistics in terms of RMSE (0.41 m) and bias (+0.19 m), results from the IC2 case bear little resemblance to the observations (Figure 3d). Significant wave heights remain within the limited range of approximately 0.60 and 0.90 m for the duration of the ice-covered season, and show little response to wind conditions during that time frame, resulting in an even higher bias than for the undamped case. WW3 simulates gravity waves, meaning that the restoring force is gravity, but for the IC2 module (and also IC3 and IC5) the restoring force is considered as the combination of gravity and the elasticity of the ice, thus modeling "flexural gravity waves." This elastic model accounts for effects of bending forces and inertia, but does not produce dissipation, so a viscosity term is introduced to produce dissipation. The continuous visco-elastic ice conditions for which the IC2, IC3, and IC5 parameterizations are intended to represent are quite different from the fractional ice cover typically observed on the Great Lakes, making other ice-wave damping schemes more suitable to the study environment.

Results from the remaining methods show comparatively poor performance. The IC3, IC4M1, IC4M2, IC4M3, IC4M5, IC4M6, IC4M7, and IC5 cases all show various degrees of overdamping (biases ranging from −0.22 m to −0.39 m). Waves simulated by these methods during the ice-covered season are generally at or near zero and only exhibit meaningful wave heights during a handful of more energetic events. This is in contrast to the more-skillful IC1 and IC4M4 configurations, both of which retain small waves in the order of 0.05–0.25 m or less during calmer portions of the winter season, consistent with the observations. As with other methods discussed, it is possible that configurations using these damping parameterizations could be improved upon through extensive testing and calibration. However, the number of different ice-wave damping schemes coupled with the number of tuning parameters make it impractical to conduct a comprehensive calibration of all methods.

The conclusions drawn from analysis of the LE4 location largely hold true at the other two 2010–2011 measurement locations, though there is some variation (Table 4). The IC4M4 method outperforms the existing IC0 simple blocking parameterization at all three locations in terms of both RMSE and bias, as does the IC1 simple damping parameterization. At all three measurement locations, the IC4M4 case has the lowest absolute bias of the ice-wave damping cases, though notably the undamped case has a lower absolute bias than any of the damped cases at the LE5 location. The undamped case also has a lower RMSE than the IC4M4 case for both LE5 and LE6 measurement locations. Interestingly, the IC2 method has the lowest RMSE of the damped cases at the LE6 location. However, the temporal coverage of in situ observations at the LE6 location is more sporadic than at the other two locations, and is generally during times of with higher waves, so comparisons at that location are less representative of conditions integrated over the winter season. Additionally, the lower cutoff limit of the AWAC

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sensor (Section 2.2) may bias the validation observational data set toward times of higher-waves, when results from the undamped case or IC2 case would better compare.

To further compare IC0 and IC4M4, cases using these parameters were plotted for all other stations and years with available in situ observations (Figure 4), which include stations LE5 and LE6 in 2010–2011 and station LE21 in 2012–2013. As with the results from LE4 in 2010–2011, the IC0 parameterization produces no wave information for much of these seasons, particularly during the higher-ice year of 2010–2011. At LE5 in 2010–2011 (Figure 4a), there are periods, such as 31 January to 4 February 2011, where the undamped model is producing waves in excess of 1 m while the in situ measurements show wave heights on the order of 0.25 m or below, and where wave heights simulated with the IC4M4 damping parameterization are reasonably consistent with those observations. Similar, though less pronounced, instances are evident at LE5 between 2–5 March 2011 and 13–14 March 2011, at LE6 on 2 February 2011 and 9–10 March 2011, and at LE21 on 1–2 March 2013 and 6 March 2013. There are also events during which the IC4M4 cases are doing well at capturing the magnitude of observed winter wave events in the presence of ice, such as LE5 between 17 and 18 March 2011, at LE6 between 6 and 7 March 2011, LE21 on 15 February 2013, LE21 on 6 March 2013, and LE21 around the time of ice offset between 12 and 16 March 2013.

However, there are other times when the undamped model captures observed events that are not represented in the damped cases, such as LE5 and LE6 between 14 and 21 February 2011, and LE21 between 19 and 20 February 2013. The presence of such instances, combined with the skill statistics showing that the undamped case has lower RMSE than damped cases at some locations (Table 4), demonstrates that while the more advanced IC4M4 ice-wave damping parameterization offers an improvement over the simple and overly aggressive IC0 parameterization currently used in the GLWUv2.0 operational forecast system, there is room for additional improvement on the configurations presented in these cases. Future work should target both improvements to the ice model, as well as identifying or generating additional winter wave observations such that the model configuration can be better calibrated to the Great Lakes ice-wave regime.

Figure 5 presents snapshots of modeled wave results for 11 January 2011 as modeled ice on Lake Erie is progressing from the western basin into the central basin of the lake (Figure 5a). The comparison of results produced by the IC0 (Figure 5b) method to those produced by the IC4M4 method (Figure 5c) demonstrates the advantage in simulation data coverage when using the IC4M4 method over the IC0 method. Wave heights simulated in the open water portions of the lake are comparable between the two cases. However, for the western and southern portions of the lake, the IC4M4 case shows damping of wave heights in those ice-covered regions, with increasingly smaller waves as distance increases from the open water. In contrast, the IC0 method blocks ice cover as land, so no wave height results are produced for those ice-covered portions of the lake. This leaves a gap in model coverage for these regions, which contain population centers where wave forecasts have particular utility. Even with imperfect damping, the wave distributions produced with the IC4M4 scheme provide forecasters with guidance they can use to generate forecasts in areas for which the current GLWUv2.0 model using the IC0 scheme would have no wave information. Human edits to the wave height and wave period forecasts to account for any ice forecast discrepancies is less time-consuming if beginning from a muted wave field compared to one in which wave heights are eliminated by the presence of ice.

Figure 6 shows visual band satellite imagery of ice cover on 9 January 2011, after the time of the model snapshot presented in Figure 5. This image shows the transitional nature of the ice regime on Lake Erie. For example, it can be seen in Figure 6 that while the western basin of the lake is mostly ice-covered, there are cracks evident through the ice sheet, and an opening is forming along the western shore of the lake as ice moves offshore. At the time shown, it is not a solid landfast sheet of ice that would be likely to result in a total damping of waves as simulated in IC0, but rather a transition ice regime that would be expected to result in a more gradual damping, consistent with the empirical IC4M4 formulation. Although Lake Erie can form landfast ice, particularly in the shallower western basin, the type of transitional ice environment depicted in this image is more common throughout most of Lake Erie, as well as the other Great Lakes (GLERL, 2023b).

In more extreme cases, there can be transitional ice cover over most or all portions of the lake. For example, on 6 March 2013, the FVCOM-CICE model showed greater than 50% ice throughout the domain (Figure 7a), which would result in no wave information across the lake under the IC0 simple blocking parameterization, thus rendering the model useless to forecasters and decision-makers. In contrast, the IC4M4 simulates a dampened wave regime under these conditions (Figure 7a), which could be used as guidance for forecasts. The ice analysis data set from the

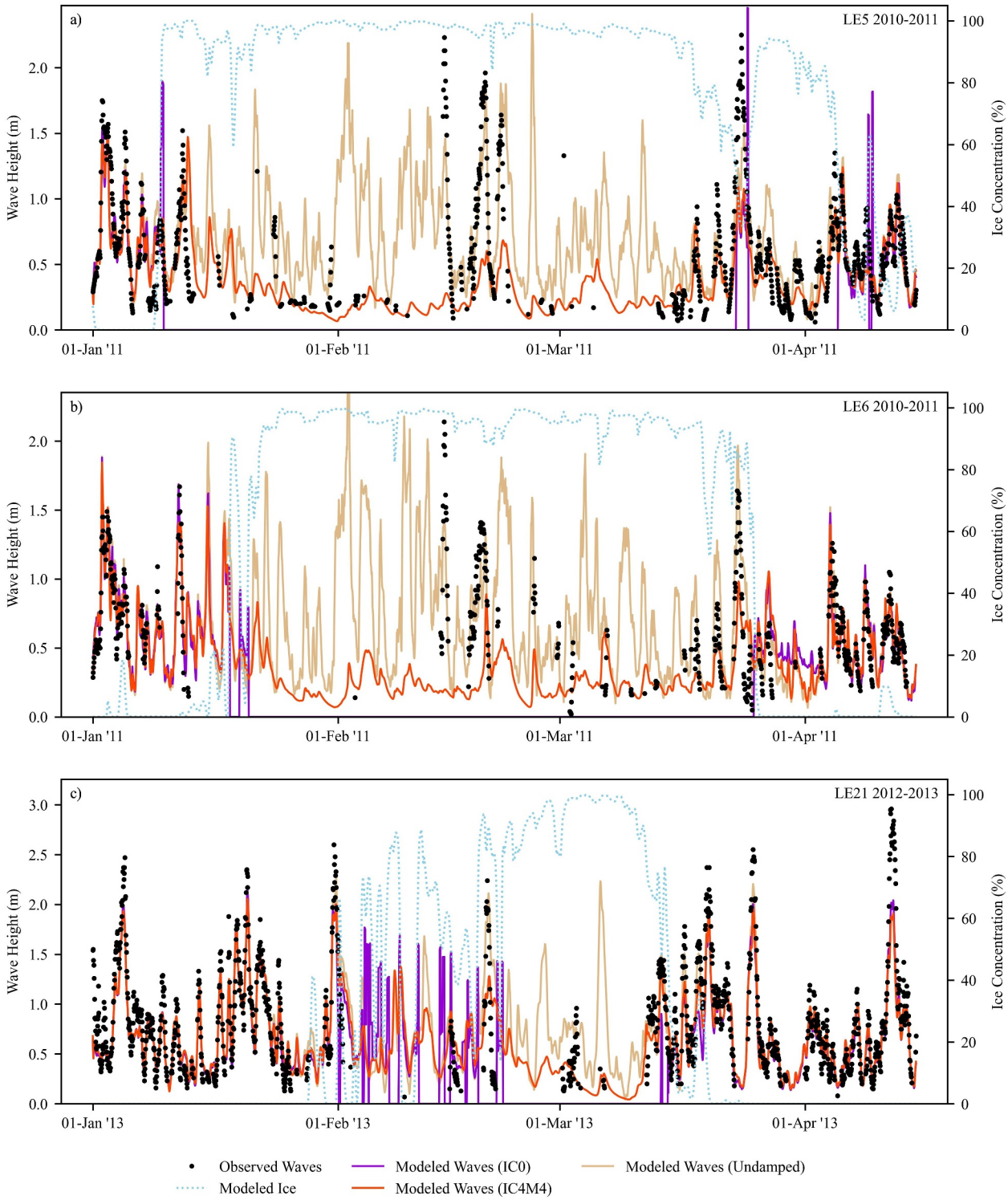


Figure 4. Comparison of modeled significant wave heights using no damping (orange line), IC0 simple ice blocking (purple line), and IC4M4 ice-wave damping (red line) to in situ observations (black dots) with modeled ice for context (dotted light blue) for panels (a) LE5 in 2010–2011, (b) LE6 in 2010–2011, and (c) LE21 in 2012-13.

same day (Figure 7b) suggests there is some degree of overicing in the FVCOM-CICE model on this day, though still shows greater than 50% ice cover for nearly all of the central basin of the lake and portions of the western and eastern basins. Satellite visual imagery from 6 March 2013 (Figure 7c) depicts a fractured and transitional ice

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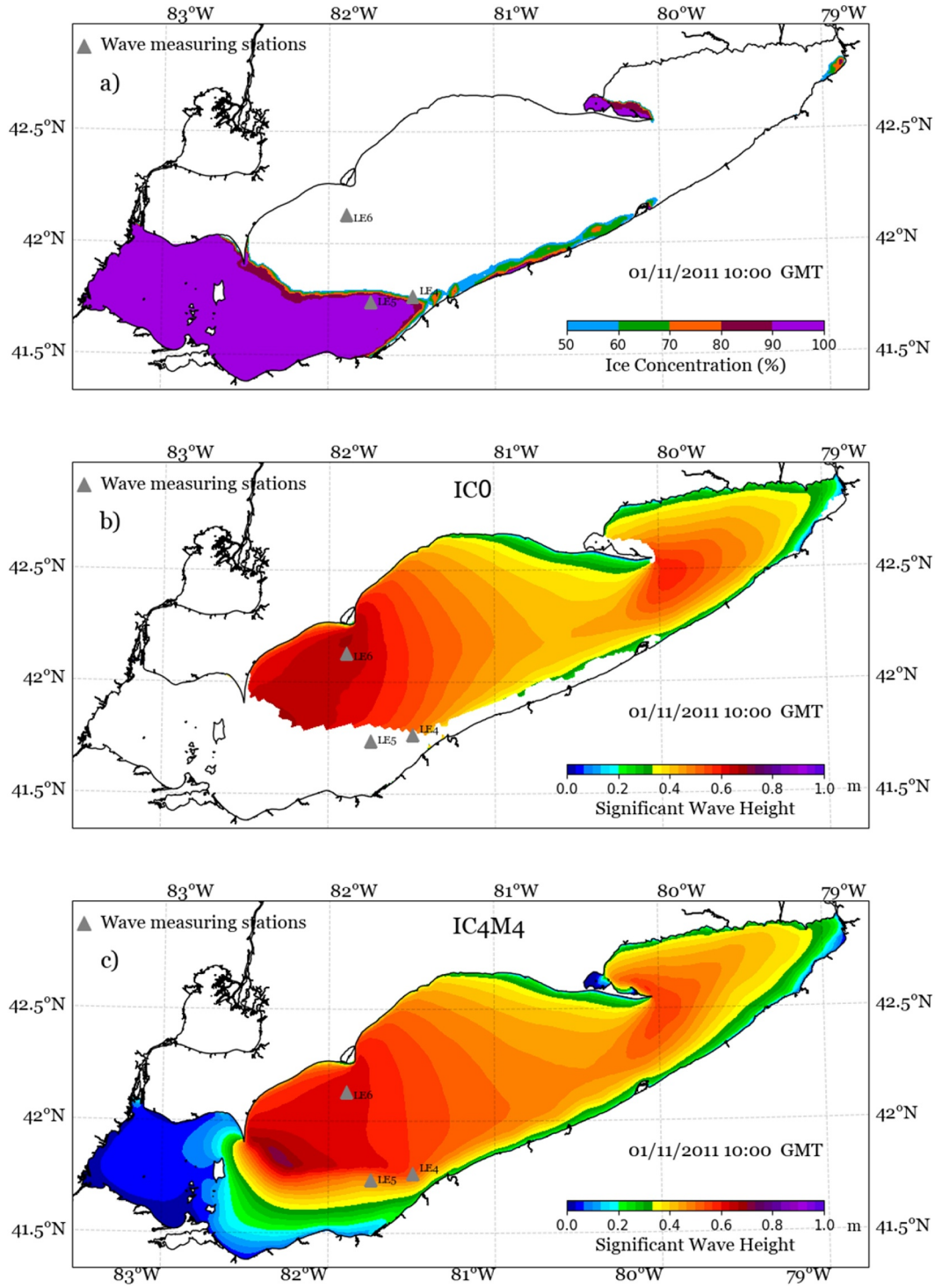


Figure 5. Domain-wide output from an example wave event on 1 January 2011 showing (a) modeled ice cover from the FVCOM-CICE simulation, (b) modeled significant wave height results from using the IC0 simple ice blocking, and (c) modeled significant wave height results using the IC4M4 ice-wave damping scheme.

environment throughout these portions of the lake that is likely to be more consistent with a damped wave environment than an environment with no waves at all. This is consistent with the comparison shown in Figure 4c, which shows observed wave heights at the LE21 location on this day ranging from 0.15 to 0.35 m, with

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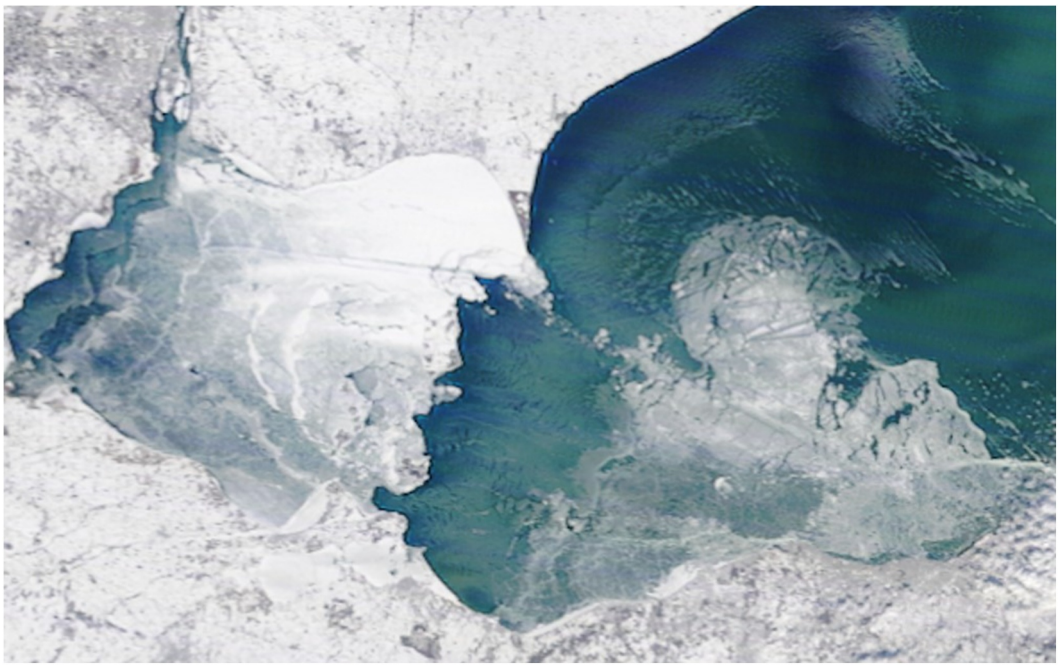


Figure 6. Moderate resolution imaging spectroradiometer (MODIS) true color satellite image, taken 9 January 2011 (NOAA Coastwatch, 2024).

corresponding results from the IC4M4 ranging from 0.20 to 0.30 m, and results from the undamped case ranging from 1.5 to 2.2 m. This demonstrates the utility during this event of the IC4M4 damped parameterization over the IC0 simple blocking scenario, and over a case that does not account for ice-wave damping effects.

The ice modules in WW3 are mostly based on the observations and studies from the Arctic Ocean and the Antarctic Ocean areas. The ice in the ocean is thicker (1–4 m) and wider when compared with Lake Erie (~0.2 m). Waves act differently in the deep open oceans than in a shallow lake. Winds of 15 m/s can generate about 5 m waves with periods about 10 s in Nordic Seas (Figure 8a, Pierson & Moskowitz, 1964), whereas it can only produce 2–2.5 m waves with periods of 5~6 s in Lake Erie (Figure 8b). The dominant wave frequency is about 0.05~0.10 Hz in the Arctic Ocean compared with 0.15~0.2 Hz in the lake (Figure 9). The wave energy spectrum ranges 0–100 m^2/s in the ocean is much higher than that in Lake Erie which ranges from 0 to 6 m^2/s. SWH response to wind speed in the deep ocean is nonlinear, possibly because of long swell waves propagation. In comparison, SWH response to wind speed trends to be linear in the shallow Lake Erie. Those formulas based on frequency may not be suitable for the shallow lakes where they cause too much damping. In the case of the ice module IC4M4, the wave attenuation is based on incoming wave height, which fits the lake—smaller waves induce smaller dampings proportionally compared with the larger wave in deep oceans.

Results from the pseudo-real-time forecast configuration are shown in Figure 10 for the 2022–2023 winter season. Although 2022–2023 was not a high-ice year on Lake Erie, periods of high ice cover are simulated intermittently throughout the winter, and wave results from these pseudo-real-time simulations are mixed (Figures 10a–10c). The IC4M4 configuration performs better overall at capturing wave dynamics through the winter season than the IC0 configuration, showing a marginal improvement in both bias and RMSE at all forecast horizons (Figure 10d). Interestingly, the skill of the wave model, in terms of both RMSE and bias, remains quite constant through the 72-hr forecast window, suggesting that the skill of both the wind forcing and ice simulations do not degrade significantly over that 3-day forecast horizon. Most notably in terms of ice-wave damping, the IC4M4 model represents, though underestimates, the moderate waves recorded on 12–14 January 2023, whereas the IC0 does not predict any waves during that period because modeled ice cover exceeds 50%. However, even using the IC4M4 scheme, many wave events throughout the winter are underestimated or missed completely. Although this suggests that the IC4M4 scheme is overdamping the waves, a variety of other factors may also be at play in this complex modeling system.

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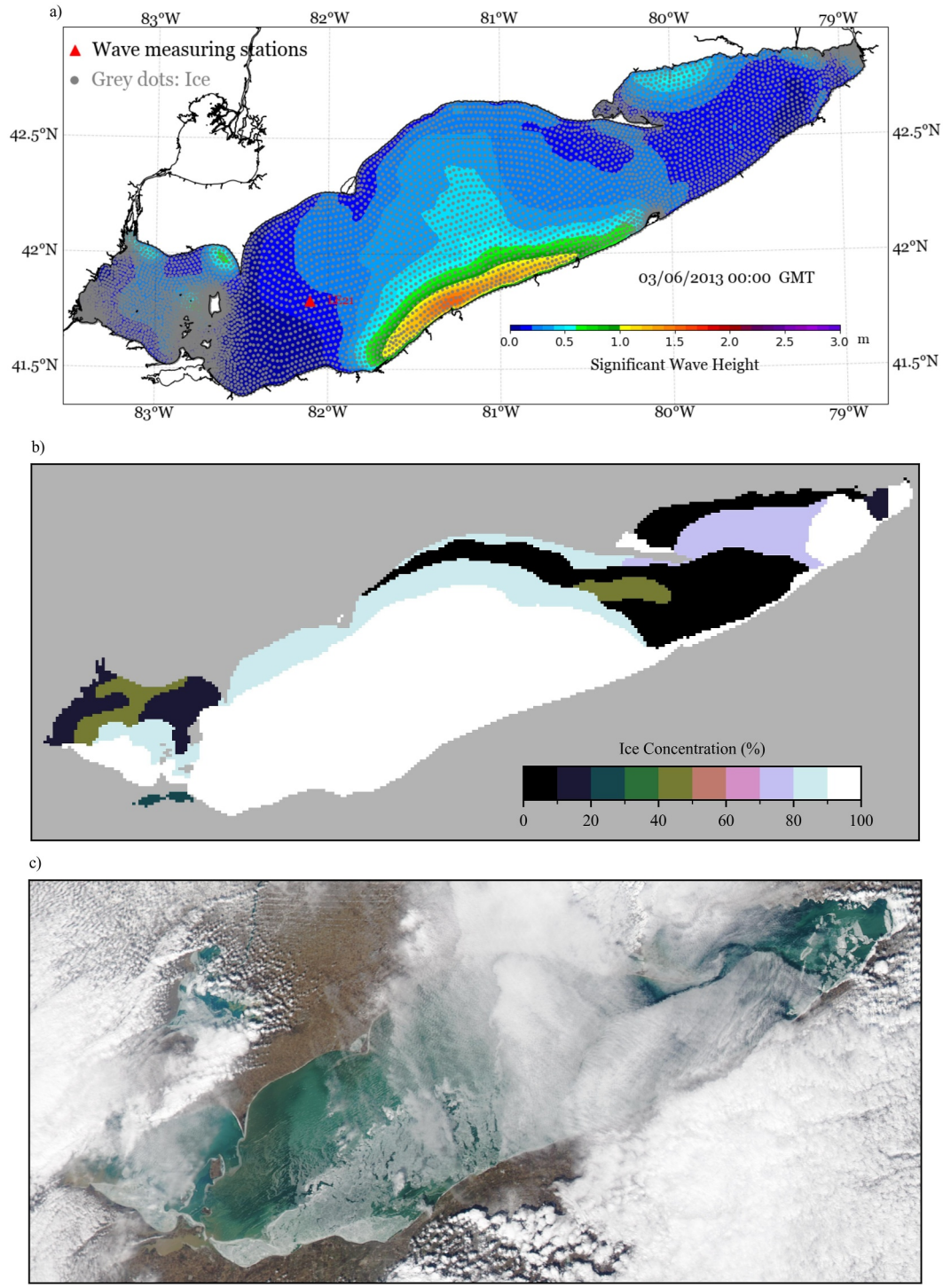


Figure 7. Domain-wide conditions on 6 March 2013 for panel (a) locations of simulated ice cover and modeled significant wave heights using the IC4M4 ice-wave damping parameterization, (b) ice coverage from the ice analysis data set, and (c) moderate resolution imaging spectroradiometer (MODIS) true color satellite image (NOAA Coastwatch, 2024).

Firstly, simulated ice cover at Reno Beach is higher than the ice analysis data set for most of the season, and while the ice analysis may not be a perfect representation of ice conditions, the consistent difference suggests that the FVCOM-CICE model is producing unrealistically high ice cover during the 2022–2023 season. In addition, the shallow nearshore location of the Reno Beach AWAC station is likely to be highly dynamic, where processes such

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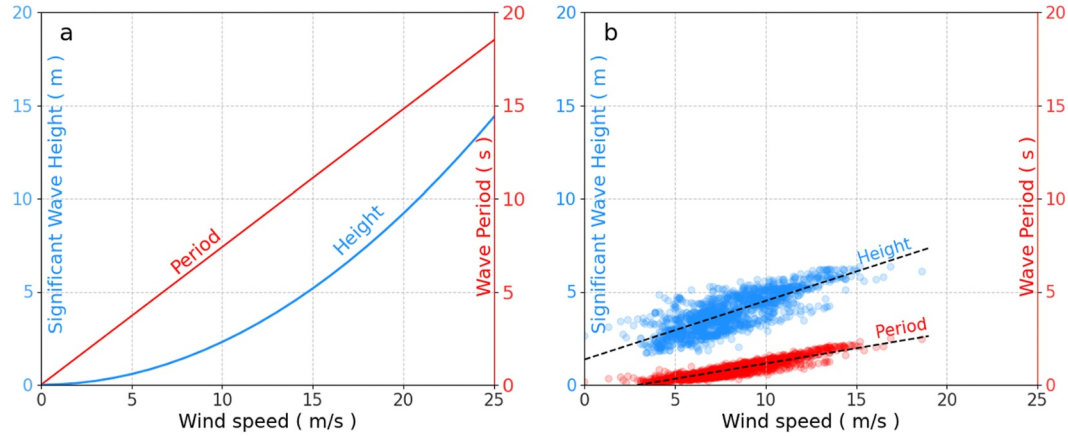


Figure 8. Significant wave height and period at the peak of the spectrum calculated from (a) of a fully developed sea in the Pierson–Moskowitz spectrum (reproduced from Pierson & Moskowitz, 1964) and (b) WW3 in Lake Erie.

as ridging are more likely to occur and offshore winds may drive ice offshore and produce open water along the shoreline. There are periods such as 25–28 December 2022 and 4–7 February 2023 where both the ice model and ice analysis show ice cover of 80%–100%, yet there are in situ-observed waves at or exceeding 1m. This suggests there are nearshore processes that may not be well-resolved in the ice model, which is designed more for the open ocean, and also that ice conditions near the shore may not be well-resolved in the ice analysis data set at the 1.8 km resolution. Third, surface wind stress in the models may be underestimated during the winter. From 20 to 25 January 2023, the Reno Beach location is shown as ice free in both the model simulations and ice analysis, yet the wave model is still underpredicting wave heights. Past studies have found a tendency for HRRR 10-m winds and wind gusts to underpredicted winds at times of higher observed speeds (Fovell & Gallagher, 2022). During the winter, when the overlake air is colder than the water at the lake surface, atmospheric conditions in the boundary layer are such that dense winds more readily reach the lake surface and transfer momentum to the lake as wind stress. As such, it is possible that the relationships used to derive wind stress at the lake surface from the standard 10-m winds used as forcing are designed for more-neutral atmospheric conditions and therefore underestimate effective wind stress for winter conditions.

Although here we focus on the development of a one-way coupled system, with waves influenced by ice, there is also feedback from the waves to the ice in the form of breakage that is not accounted for in this study. Specifically, waves can break ice and increase open leads. It has been found that ice break-up in the marginal ice zone can generate lognormal floe size distributions (Mokus & Montiel, 2022). The development of leads and changes in

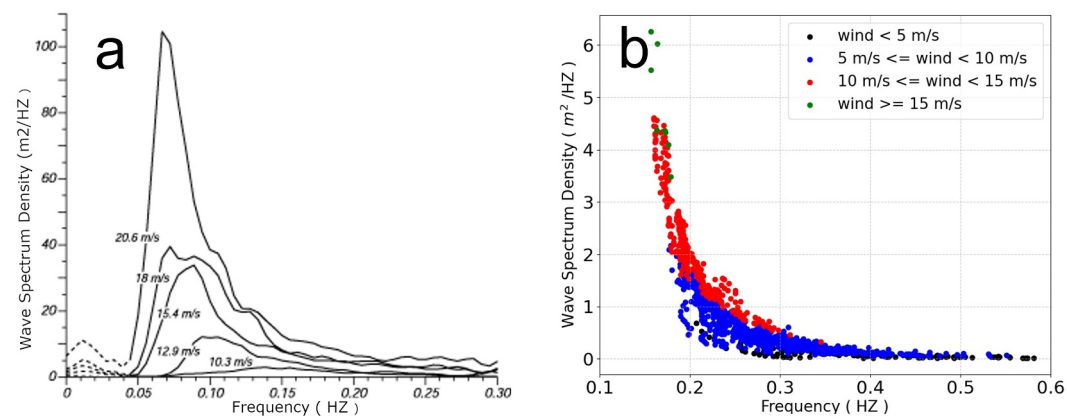


Figure 9. Wave spectra for different wind speeds from (a) Moskowitz, 1964 of a fully developed sea and (b) WW3 in Lake Erie.

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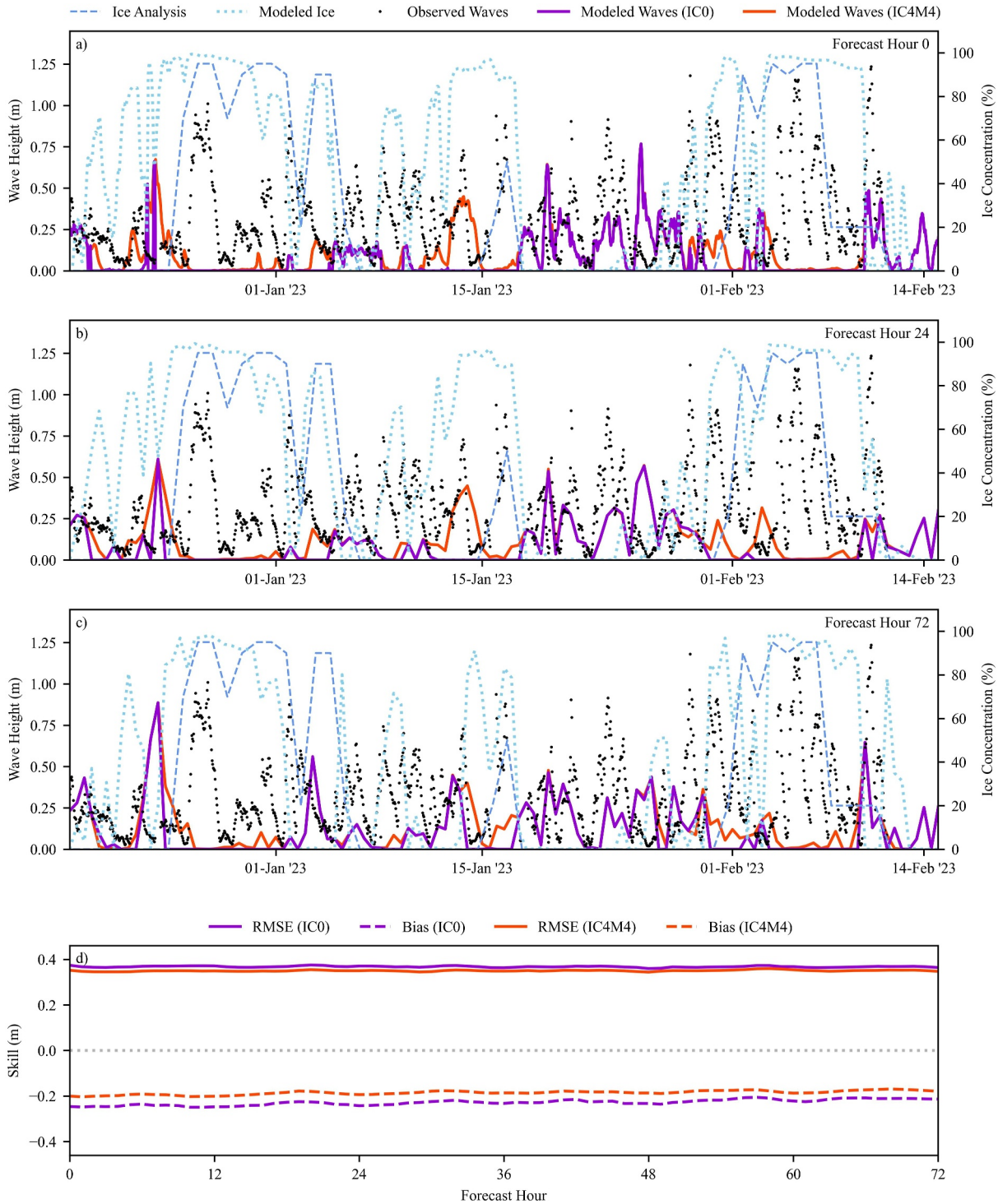


Figure 10. Forecast skill assessment for the 2022–2023 season comparing modeled significant wave heights using IC0 simple blocking (purple line) and IC4M4 ice-wave damping (red line) to in situ observations at Reno Beach station (black dots) with simulated ice (dotted light blue) and ice analysis (dotted dark blue) for context at panel (a) forecast hour 0, (b) forecast hour 24, and (c) forecast hour 72, as well as panel (d) Significant wave height skill statistics root mean square error and bias as a function of forecast hour for the period of 18 December 2022–15 February 2023.

floes size will alter the rate of ice formation or melting within the coupled air-ice-water system. The significance of these wave-induced ice breakage processes on winter wave heights is not quantified here, and remains an area of future research in the Great Lakes.

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The fractional ice regime of Lake Erie, and the other Great Lakes, makes it exceptionally difficult to collect in situ observations during the winter season, and it is well-documented that there is a bias in observational data sets toward ice free conditions (Ozersky et al., 2021). Although we have identified possible areas for improving the coupled ice-wave modeling system, including the ice-wave damping parameterization, the ice model, the wave model wind stress parameterization, and addition of wave-induced ice breakage, additional observations will be critical toward characterizing the importance of these mechanisms and further calibrating the modeling system.

4. Conclusion

WAVEWATCH III (WW3, version 6.07.1) with ice damping modules were tested in Lake Erie in a loosely coupled ice-wave modeling system, and the results were compared with in situ moored AWAC wave observations. The comparisons show that the empirical ice-wave damping module IC4M4 generally demonstrated the best performance among those tested, and consistently outperforms the IC0 simple blocking scheme currently used in the operational GLWUv2.0 system for both hindcast and forecast configurations. However, the model may further benefit from additional research and development to improve ice simulations particularly during the spring melting period, further calibration of the IC4M4 ice-wave damping module, and the momentum transfer of wind to the lake surface during the winter. Collection of winter observations on Lake Erie are challenging, because of its marginal ice regime, but will be critical toward further calibration and validation of the coupled ice-wave modeling system. The ice modules in WW3 are mostly based on observations and studies from the Arctic Ocean and the Antarctic Ocean. The peak wave frequency is about 0.05∼0.10 Hz in the Arctic Ocean compared to 0.15∼0.2 Hz in the lake. Formulas based on frequency are not suitable for the shallow lakes where they cause too much damping. With the ice module IC4M4, although the module is also based on the Antarctic Ocean study, the wave attenuation is based on incoming wave height that also fits the shallow lake.

Data Availability Statement

WAVEWATCH III model is publicly accessible at https://github.com/NOAA-EMC/WW3. The Reno Beach wave data is publicly accessible at https://www.glerl.noaa.gov/res/recon/station-rbs.html. The buoy data has been submitted to National Centers for Environmental Information (NCEI) for public release.

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